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University of Asia Pacific

Department of Computer Science and Engineering

CSE 438: Robotics Lab

LAB REPORT

Experiment Number: 03

Experiment Title: Design an obstacle avoidance maze-solving robot.

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1. Experiment Name

Designing an Obstacle Avoidance Maze-Solving Robot Using Arduino UNO and Ultrasonic Sensor

2. Objective

The main objective of this experiment is to design and build an autonomous robot that can move through a maze while avoiding obstacles on its own. The robot uses an **HC-SR04 ultrasonic sensor** to detect obstacles and measure the distance in front of it. Based on the detected distance, the **Arduino UNO** controls the two DC motors through the **L293D motor driver**, allowing the robot to move forward, stop, reverse, and turn when necessary.

3. Apparatus / Hardware & Software Requirements

- List all required tools and components:

Hardware Requirements:

- Arduino UNO
- L293D Motor Driver IC
- 2 × DC Gear Motors
- Breadboard
- Jumper Wires
- 9V Battery / External Power Supply
- Robot Chassis
- Wheels

Software Requirements

- Arduino IDE
- Tinkercad Circuits (for simulation/design)
- Arduino programming language (C/C++)

4. Circuit Diagram / Schematic

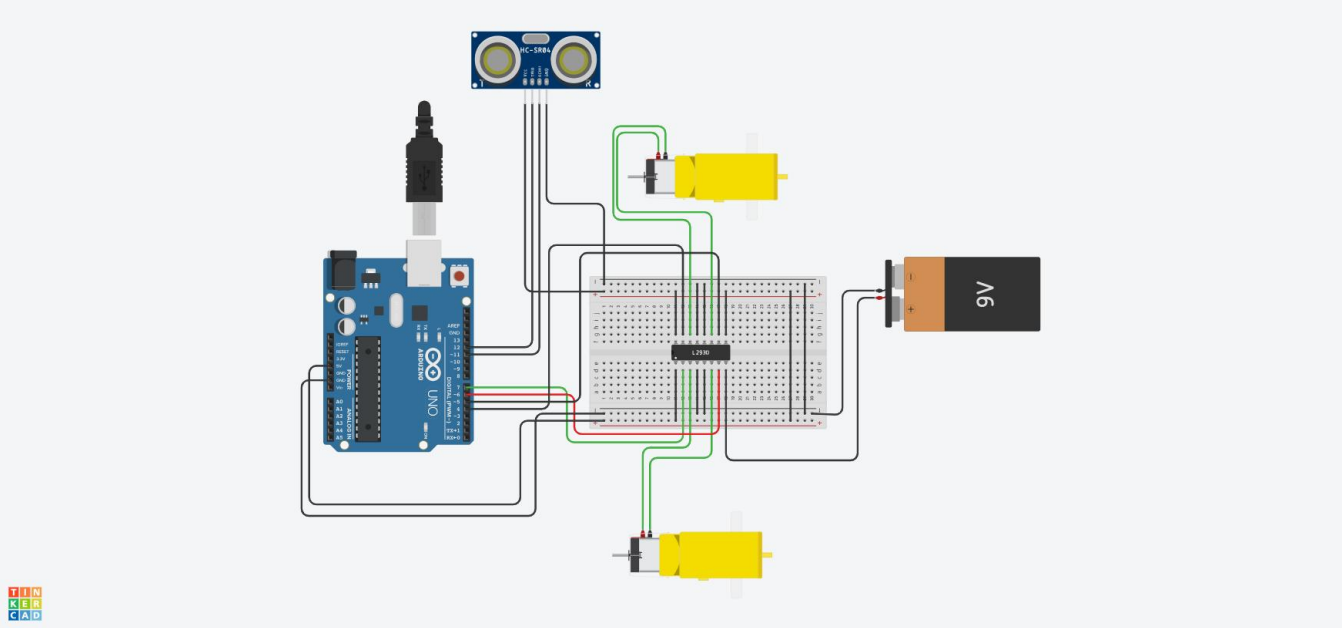


Fig: obstacle avoidance maze-solving robot

5. Code / Assembly Program

```
#include <Servo.h>

//Obstacle Avoidance Robot

const int trig=12;
const int echo=11;
const int leftFrwrd=7;
const int leftBack=6;
const int rightFrwrd=5;
const int rightBack=4;

int duration=0;
int distance=0;

void setup() {
  pinMode(trig,OUTPUT);
  pinMode(echo,INPUT);
  pinMode(leftFrwrd,OUTPUT);
  pinMode(leftBack,OUTPUT);
  pinMode(rightFrwrd,OUTPUT);
  pinMode(rightBack,OUTPUT);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(trig,LOW);
  delayMicroseconds(2);
  digitalWrite(trig,HIGH);
  delayMicroseconds(10);
  digitalWrite(trig,LOW);
  duration=pulseIn(echo,HIGH);
  distance=(duration*0.034/2);
  Serial.print("Distance: ");
  Serial.println(distance);

  if(distance<20)
  {
    digitalWrite(leftFrwrd,LOW);
    digitalWrite(leftBack,HIGH);
    digitalWrite(rightFrwrd,HIGH);
    digitalWrite(rightBack,LOW);
    delay(100);
  }

  else
  {
```

```
  digitalWrite(leftFrwrd,HIGH);
  digitalWrite(leftBack,LOW);
  digitalWrite(rightFrwrd,HIGH);
  digitalWrite(rightBack,LOW);
 }
}
```

6. Output / Observations

After uploading the program to the Arduino, the robot behaves according to the distance measured by the HC-SR04 ultrasonic sensor.

Observed Behavior:

Ultrasonic Distance	Robot Action
Greater than 20 cm	Robot moves forward
20 cm or less	Stops
Obstacle detected	Moves backward
After reversing	Turns right
No obstacle	Continues forward

When the **center flame sensor detects fire**, the robot stops moving and the water pump starts spraying water. The buzzer also turns on to indicate that fire has been detected.

Working Procedure

1. The HC-SR04 sensor continuously sends ultrasonic waves.
2. The sensor receives the reflected wave from an obstacle.
3. Arduino calculates the distance from the obstacle.
4. If the distance is greater than **20 cm**, both motors rotate forward.
5. If an obstacle is detected within **20 cm**, the robot stops.
6. The robot moves backward for a short time.
7. It then turns right to find another path.
8. The process continues automatically.

7. Result

The obstacle avoidance maze-solving robot was successfully designed and implemented using an **Arduino UNO, HC-SR04 ultrasonic sensor, L293D motor driver, and two DC motors**. The robot was able to detect obstacles in its path and change its direction automatically based on the detected distance.

The robot performed the required movements without manual control. **Therefore, the objective of the experiment was successfully achieved.**

8. Conclusion

In this experiment, we successfully designed and developed an obstacle avoidance robot using an **Arduino UNO, HC-SR04 ultrasonic sensor, L293D motor driver, and two DC motors**. The ultrasonic sensor helped the robot detect obstacles in its path, while the Arduino controlled the motors according to the detected distance.

Through this project, we learned how to connect and use an ultrasonic sensor with Arduino, measure the distance of nearby objects, and control DC motors using the L293D motor driver. We also gained practical knowledge about how an autonomous robot can make decisions and change its movement based on its surroundings.

Overall, this experiment gave us a better understanding of **obstacle detection, motor control, robot navigation, and autonomous robotic systems**.